



Ecole Polytechnique

Computer Graphics

Assignment 2

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1 Lab 6: Voronoi diagrams

1.1 Computing a Voronoi diagram

The first step consists of computing a two-dimensional Voronoi diagram in the unit square $[0, 1]^2$. In this example, the diagram is generated from 4 sites with coordinates $(0.1, 0.2)$, $(0.6, 0.4)$, $(0.5, 0.7)$, and $(0.2, 0.5)$. Each site defines one Voronoi cell, containing all points that are closer to this site than to any other site.

The cells are computed using Voronoi Parallel Linear Enumeration. For each site, the cell is initialized as the unit square and is then clipped successively by the perpendicular bisectors with the other sites. The Voronoi computation itself took 0.023, ms.

1.2 Sutherland–Hodgman polygon clipping

The clipping step is implemented using the Sutherland–Hodgman polygon clipping algorithm. For each edge of the current polygon, the algorithm checks whether the two endpoints are inside or outside the half-plane by comparing squared distances:

$$|X - P_0|^2 \leq |X - P_i|^2,$$

where P_0 is the current site and P_i is another site. If an edge crosses the bisector, the intersection point is computed and added to the new polygon.

The total execution time was 205.023, ms.

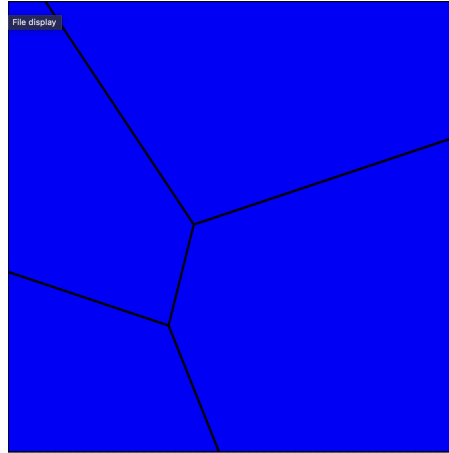


Figure 1: Voronoi diagram computed from four sites using linear enumeration and Sutherland–Hodgman clipping.

2 Lab 7: Power diagram and L-BFGS weight optimization

The second step extends the Voronoi diagram to a Power diagram. In this test, $N = 100$ sites are generated randomly in the unit square $[0, 1]^2$ using a fixed random seed equal to 42. All weights are initialized to 0, and then optimized using the L-BFGS algorithm.

For a site P_i with weight w_i , the cell is computed using the power distance

$$|X - P_i|^2 - w_i.$$

Compared to the standard Voronoi diagram, the separating line between two sites is shifted according to the difference between their weights. In the implementation, this is done, where the midpoint of the segment is replaced by a weighted midpoint. At each L-BFGS iteration, the Power diagram is

recomputed, the area of each cell is measured, and the gradient is given by the difference between the current cell area and the target area. Since the sites are uniformly weighted, the target area for each cell is

$$\lambda_i = \frac{1}{N}.$$

After optimization, the final cells are saved both as a PNG image and as an SVG file. Figure 2 shows the two generated outputs.

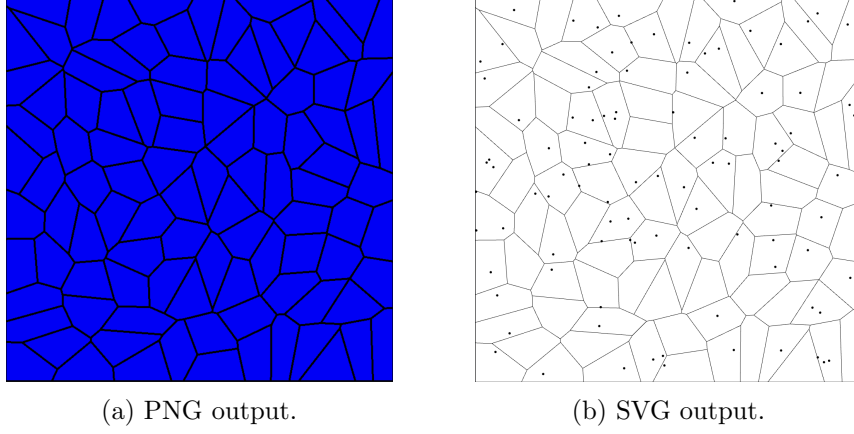


Figure 2: Power diagram computed from 100 random sites after L-BFGS weight optimization.

3 Lab 8: Free-surface fluid simulation

The final step consists of implementing a two-dimensional free-surface fluid simulation using the Gallou”et–M’erigot incompressible Euler scheme. In this test, the simulation is initialized with 300 particles. The particles are generated randomly with seed 42, with initial positions in the rectangle $[0.15, 0.45] \times [0.10, 0.80]$, and all initial velocities are set to zero.

At each time step, the particle positions are used as sites of a Power diagram. The optimization is performed with L-BFGS in order to prescribe the fluid cell areas. The total fluid volume is set to 0.25, so the target area of each particle is

$$\frac{0.25}{300}.$$

The partial optimal transport formulation also includes an air weight. Each Laguerre cell is intersected with a disk, approximated using 48 polygon edges.

After the optimized cells are computed, the centroid of each cell is used to apply a spring force to the corresponding particle:

$$F_i = \frac{c_i - x_i}{\varepsilon^2} + m_i g,$$

where x_i is the particle position, c_i is the centroid of its Laguerre cell, $\varepsilon = 0.004$, $m_i = 200$, and $g = (0, -9.81)$. The time step is set to $dt = 0.002$, and the simulation is run for 1000 frames. The frames generated by the simulation were also assembled into a video. The final video file is named `fluid.mp4` and is included in the GitHub repository together with the source code and report

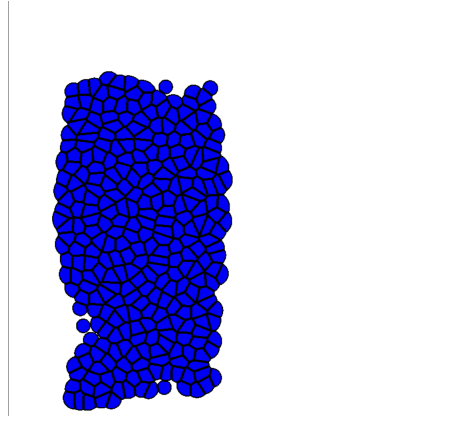


Figure 3: Frame from the free-surface fluid simulation with 300 particles.

4 Comments

The code was developed progressively across the different lab sessions. The first step was to implement polygon clipping and Voronoi cells. Then, the same structure was extended to weighted Power diagrams. After that, the weights were optimized with L-BFGS in order to impose target cell areas. Finally, these components were combined to implement the two-dimensional incompressible fluid simulation.

This project was completed individually. I shared opinions and ideas with my classmates: Sacha Gregoire, Youssef Seghrouchni, Ayoub and Rachid.

5 Conclusion

This assignment implemented the main components needed for a two-dimensional free-surface fluid solver. Starting from Voronoi diagrams and Sutherland–Hodgman clipping, the project was extended to Power diagrams and L-BFGS weight optimization. These tools were then used in the Gallou”et–M’erigot incompressible Euler scheme to simulate a fluid inside a disk.